Practical No.2

Title: Write a program to implement Huffman Encoding using a greedy strategy.

```
import heapq
# Node class for Huffman Tree
class Node:
  def __init__(self, freq, char=None, left=None, right=None):
    self.freq = freq
    self.char = char
    self.left = left
    self.right = right

  # Compare nodes based on frequency
  def __lt__(self, other):
    return self.freq < other.freq

# Function to generate Huffman Codes
def generate_codes(node, code="", huffman_code={}):
  if node is None:
    return

  # If leaf node
  if node.char is not None:
    huffman_code[node.char] = code

  generate_codes(node.left, code + "0", huffman_code)
  generate_codes(node.right, code + "1", huffman_code)

  return huffman_code

# Character frequencies
chars = {
  'A': 6,
  'B': 7,
  'C': 2,
  'D': 23,
  'E': 22,
  'F': 47
}

# Create priority queue
heap = []

for char, freq in chars.items():
  heapq.heappush(heap, Node(freq, char))

# Build Huffman Tree
while len(heap) > 1:
  left = heapq.heappop(heap)
  right = heapq.heappop(heap)

  merged = Node(left.freq + right.freq, left=left, right=right)

  heapq.heappush(heap, merged)

# Root node
root = heap[0]

# Generate Huffman Codes
codes = generate_codes(root)
```

```
print("Huffman Codes")
print("---------------------------")

for char in sorted(codes):
   print(char, ":", codes[char])

# Calculate Average Code Length
total_frequency = sum(chars.values())

average = 0

print("\nEncoded Length of Each Character")
print("-----------------------------------")

for char in sorted(chars):
   bits = len(codes[char]) * chars[char]
   average += bits
   print(char, "Frequency =", chars[char],
       "Code =", codes[char],
       "Bits =", bits)

print("\nAverage Code Length =", round(average / total_frequency, 2), "bits")

print("Total Huffman Encoded Message Length =", average, "bits")
```

```
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code> & C:\Users\HP\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/HP/Downloads/2_Huffman.py
---------------------------
A : 11011
B : 1100
C : 11010
D : 10
E : 111
F : 0

Encoded Length of Each Character
-----------------------------------
A Frequency = 6 Code = 11011 Bits = 30
B Frequency = 7 Code = 1100 Bits = 28
C Frequency = 2 Code = 11010 Bits = 10
D Frequency = 23 Code = 10 Bits = 46
E Frequency = 22 Code = 111 Bits = 66
F Frequency = 47 Code = 0 Bits = 47

Average Code Length = 2.12 bits
Total Huffman Encoded Message Length = 227 bits
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code>
```